Nooelec UWB Surveyor: apparent distance offset Measurement details and complex S_{11} data

Technical note for John

9 October 2026

Dear John,

Thank you for your help. This note documents the instrument, time-domain processing and measurement geometry, and accompanies complex S_{11} files for a target-free background and three plate positions. The purpose is to distinguish a common electrical delay from the change in free-space distance, without assuming that the antenna alone explains the offset.

1 Equipment and geometry

- Instrument: LiteVNA 64, operated via USB using custom Python scripts.
- Antenna: Nooelec UWB Surveyor, planar biconical PCB antenna, 120 × 120 mm, marked v1.0b2 and 2025-10-15.
- Target: an aluminum plate, parallel to the antenna PCB. Distances are measured from the antenna's front surface to the plate and are approximate.
- **Plate width, height and thickness: not yet supplied.**
- The apparatus is used indoors on a table, with wooden supports and nearby fixtures. No anechoic environment or multipath suppression is claimed.

The transparent panel visible in a previously shared setup photograph was absent during the aluminum-plate measurements.



Figure 1: Antenna, support, cable routing and VNA. This is an AI-retouched presentation image: household surroundings and a person were removed. It is not evidence of an uncluttered measurement environment or a dimensional reference. The plate is not shown.

2 Acquisition and processing

Frequency band	1–6 GHz
Frequency samples	201, uniformly spaced
Frequency step	25 MHz
Plate averaging	10 sweeps, complex arithmetic mean
Calibration	Software one-port SOL at the cable end
Window	Symmetric Hann, $w_k = \frac{1}{2}[1 - \cos(2\pi k/(N-1))]$
IFFT length	$16N = 3216$ (zero-padding factor 16)
Reference impedance	$50\,\Omega$

The firmware is read in raw mode, and the saved SOL coefficients are applied in Python. The antenna is connected after the calibration plane; its radiating transfer function is not de-embedded. The exported complex values have not been windowed or background-subtracted.

For each plate position, the displayed method-A profile uses

$$R_k = \overline{S}_{11,\mathrm{plate}}(f_k) - \overline{S}_{11,\mathrm{background}}(f_k), \tag{1}$$

$$P_m = 16\,\left|\mathrm{IFFT}_{3216}(w_k R_k)_m\right|, \tag{2}$$

$$d_m = \frac{c}{2}\frac{m}{3216\,\Delta f}, \qquad c = 299792458 \text{ m/s}. \tag{3}$$

The reported level is $20\log_{10} P_m$. The Hann coherent gain is not normalized out. The frequency start contributes a phase factor to the complex time response but does not change its magnitude envelope. The plotted positive-distance interval is approximately 0–3 m. The interpolated distance spacing is 1.86 mm; this is *not* the physical resolution. The bandwidth scale $c/(2B)$ is about 3 cm, and Hann broadens the lobes.

3 Data supplied: 9 October 2026

The currently available blue-antenna dataset contains three positions at approximately 50, 60 and 70 cm. All three files have exactly the same frequency grid, background and SOL coefficients; these were checked numerically. The mapping between filenames and distances is recorded in the project's existing comparison manifest.

Measurement	UTC acquisition timestamp	Export basename
Background	15:26:25	`background`
Plate, 50 cm	15:27:14	`plate_50cm`
Plate, 60 cm	15:27:44	`plate_60cm`
Plate, 70 cm	15:27:56	`plate_70cm`
Independent SHORT	10:09:10	`independent_short`

Each basename is provided as CSV and Touchstone `.s1p` in `data/`. CSV columns are frequency in Hz, real S_{11}, imaginary S_{11}. Touchstone files use `# Hz S RI R 50`. These are calibrated complex means *before* background subtraction. The SHORT uses the same SOL coefficients but was acquired earlier in the day.

4 Plate-position comparison

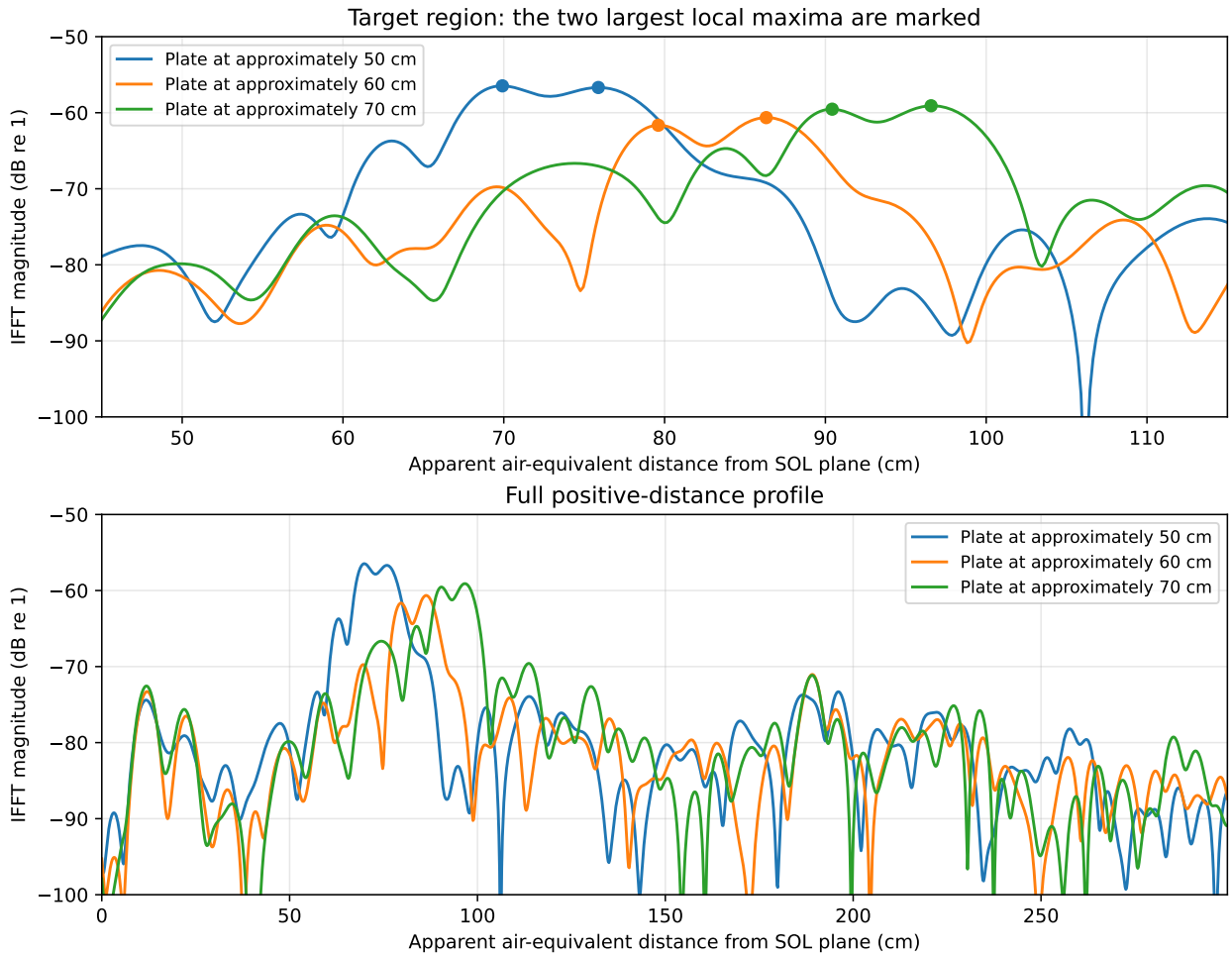


Figure 2: 9 October measurements processed with identical background subtraction, Hann window and IFFT settings. No distance-offset correction or per-curve amplitude normalization has been applied. The dots mark the two strongest local maxima in each profile, without assigning them to physical interfaces.

Plate (cm)	Near peak (cm)	Far peak (cm)	Near offset (cm)	Far offset (cm)
50	69.91	75.88	19.91	25.88
60	79.61	86.32	19.61	26.32
70	90.42	96.57	20.42	26.57

The near-peak shifts are 9.69 and 10.81 cm for nominal 10 cm movements; the far-peak shifts are 10.44 and 10.25 cm. The mean offsets are approximately 20.0 cm and 26.3 cm for the near and far branches, respectively. The strongest peak changes branch between 50 and 60 cm, so simply selecting the global maximum introduces a jump unrelated to the nominal displacement. The paired peaks do not establish two physical target surfaces.

5 Common echo-phase slope and deviations

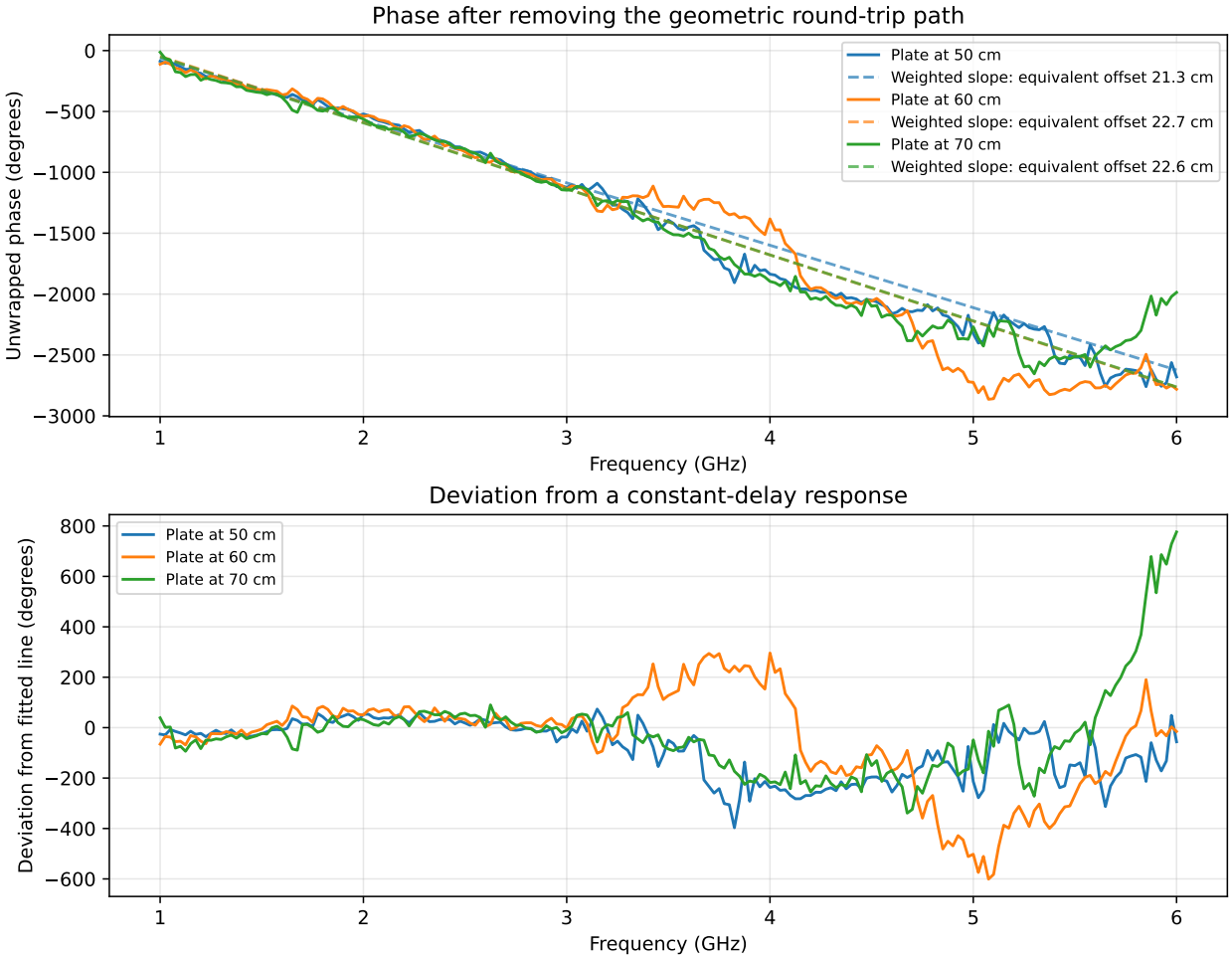


Figure 3: English reproduction of the middle and bottom panels of `phase_echo_bleue_50_60_70cm.png`. Top: phase after removing the nominal geometric round-trip path, with independent power-weighted linear fits. Bottom: deviation from each fitted line. These are plate-minus-background echoes measured with the blue antenna, not the antenna-only S_{11}.

The processing order is important: first remove the nominal free-space phase from the complex echo, then unwrap its phase:

$$H_d(f) = [\overline{S}_{11,d}(f) - \overline{S}_{11,\mathrm{bg}}(f)]e^{+\mathrm{j}4\pi fd/c}, \qquad \phi_d(f) = \mathrm{unwrap}\arg H_d(f).$$

Each straight line is fitted over 1–6 GHz with weights proportional to $|H_d(f)|^2$ and a free intercept. Its slope b corresponds to $\tau = -b/(2\pi)$ and an equivalent offset $c\tau/2$. The offsets are approximately **21.3, 22.7 and 22.6 cm** for the 50, 60 and 70 cm positions. Their similar slopes support a common delay component after nominal free-space propagation has been removed.

The lower panel shows departures from a constant-delay response, particularly above approximately 3 GHz, where the echo magnitude also decreases. These departures can include antenna dispersion, target/fixture multipath, noise and phase-unwrapping errors near spectral minima. Weighting reduces the influence of low-amplitude samples but cannot repair an unwrapping error. The common slope alone does not uniquely identify the antenna radiation delay.

6 Earlier observations, kept separate

On 22 September, four approximate positions of 50, 60, 70 and 80 cm gave near peaks of 68.4, 81.3, 91.7 and 100.9 cm, respectively, corresponding to an average near-peak offset of about 20.6 cm. Those values are retained from the earlier analysis in our correspondence. The original September acquisition files are no longer present in the current workspace, so they are *not* the data exported with this note. The October dataset above independently shows a similar near-peak offset. An earlier experiment with a different, green antenna reportedly showed an offset of approximately 38 cm. Its raw data and complete configuration are not included, so it is not used here as a controlled antenna comparison.

7 Independent SHORT check and questions

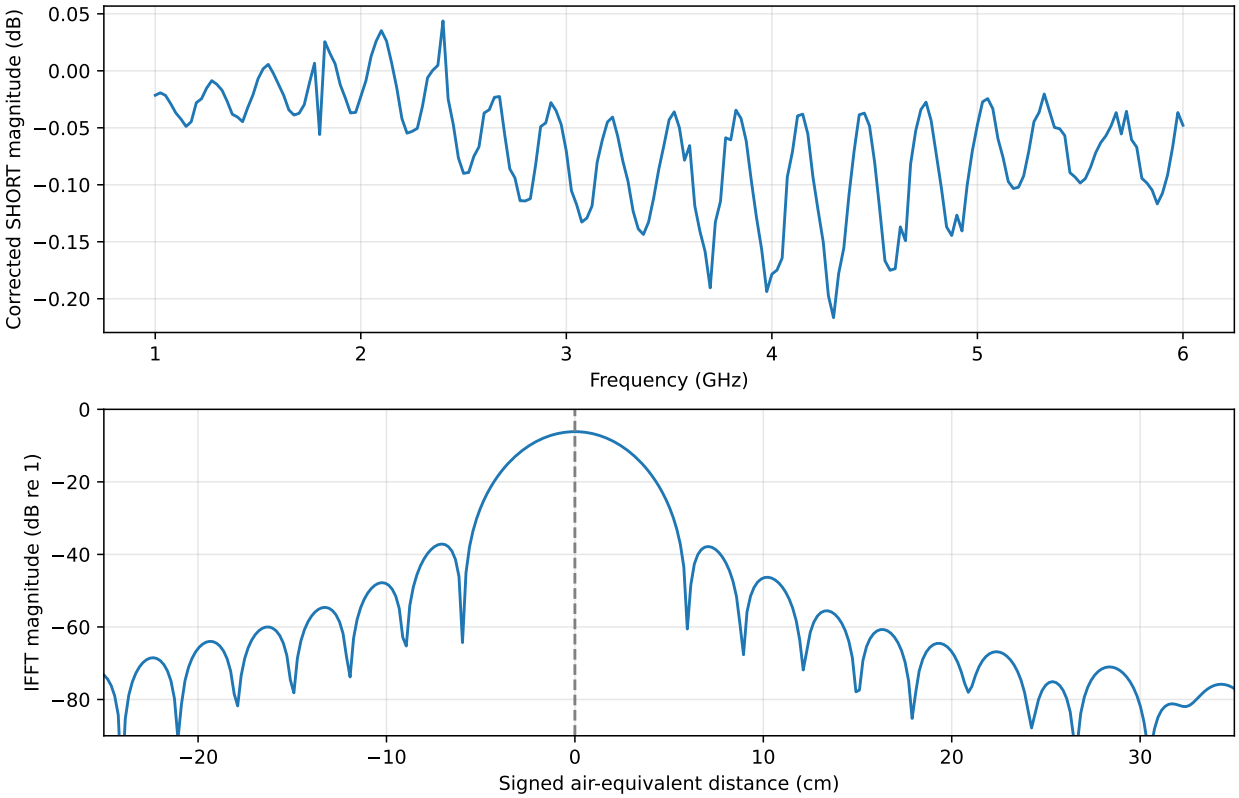


Figure 4: A new SHORT acquisition on 9 October, corrected with the existing SOL rather than used to fit it. The signed distance axis shows both sides of zero. The SHORT profile is computed directly, without background subtraction.

This check tests the calibrated chain at the cable end. It does not measure the antenna's radiation delay and does not rule out later drift, reconnecting effects or room multipath. Its zero-distance response supports investigating contributions introduced after the calibration plane.

A 21 cm apparent offset corresponds to a total round-trip delay of about 1.4 ns. We have not established a physical decomposition of that delay. Could you advise on the following points?

1. Are group-delay or complex radiation-response data available for the UWB Surveyor v1.0b2 over 1–6 GHz, referenced to the SMA connector?

2. Is its phase-center location versus frequency and observation direction characterized, and what reference convention is used?

3. Could the feed structure and radiating response account for the common near-peak offset, or for the paired-lobe profile? What measurement would best distinguish these effects from fixture or room reflections?

Manufacturer reference: UWB Surveyor datasheet, revision 1 (2022). The published sheet does not specify group delay or phase-center coordinates; it predates the marking on this antenna.